

Eco Bénin Plast

Digital Plastic Waste Management Platform

01 PROBLEM

Context & Challenge

Eco Bénin Plast is a Beninese startup founded by Servane Kpatoukpa in 2024, with a team of 3 co-founders. The platform digitally connects plastic waste-producing households with transformation companies (recycling, valorisation), currently piloted in Ouidah, 30 km from Cotonou.

Core Problem Statement

Only 7–14% of plastic waste is valorised globally. In Benin, collection is uneven and limited to Cotonou. Transformation companies struggle to source raw materials consistently while households in peri-urban zones have no structured waste valorisation outlet.

Identified UX Problems

01 Multi-Profile Onboarding Undefined

2 user types (households and companies) with very different needs, but no distinct entry paths on the platform - creating immediate confusion.

02 Unclear Economic Model

Who pays what? How are households compensated, at what rate? Who manages financial flows? These questions were unresolved at the start of the UX work.

03 Operationally Overloaded Model

Initial concept included transport, physical collection points, and logistics - incompatible with a 3-person startup's resources and budget.

04 Uncertain Household Adoption

Households must understand the benefit of registering, sorting waste, and reporting availability - a major educational effort that wasn't yet planned.

Methodology

Research combined founder coaching sessions (Programme Mastercard Foundation), competitive analysis across 4 actor types, and persona development from real prospected user profiles (50 households prospected, 40 positive responses).

UX Competitive Analysis

UX Criterion	SGDS (State)	Local NGOs	Physical Model	Eco Bénin Plast
Geographic Coverage	Cotonou only	Occasional	Limited	National (target)
Digital Accessibility	None	Low	None	Web + USSD - Differentiator
Household Remuneration	✗	✗	✗	✓ Differentiator
Enterprise Matching	✗	✗	Manual	✓ Automated
Traceability / History	✗	✗	✗	✓ To build
Scalable Model	✗	✗	✗	✓ Lightweight marketplace
Rating / Trust System	✗	✗	✗	✓ To build

✓✓ = *Very good*

✓ = *Present*

✗ = *Absent*

User Personas

Adjoua Yarou - Household Collector Age: 38 · Peri-urban zone, near Ouidah Goal: Earn supplementary income, contribute to a cleaner environment Frustration: No regular collection in her neighbourhood. No reliable internet access. UX Need: Simple interface (ideally USSD), understand registration, know earnings and timing	M. Agossou - Transformation Company Role: Manager of a recycled goods SME, Cotonou · 8 employees Goal: Obtain plastic raw materials regularly at lower cost Frustration: Must procure from the state entity or landfill sites - slow and unreliable. UX Need: See availability by zone, quantities, place orders, pay directly on the platform
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Key UX Insights

Insight #1 - USSD access is a non-negotiable differentiator

For peri-urban households without reliable internet, USSD is the only viable entry point. Any web-only solution immediately excludes the core target user.

Insight #2 - Profile separation must come first

Household and enterprise users have completely opposed needs. Conflating them on a single interface creates immediate confusion and abandonment from the first screen.

Insight #3 - Household remuneration must be visible before sign-up

The key motivator for household participation is earning money. If the compensation model is unclear at onboarding, registration rates collapse.

Insight #4 - Trust requires a mutual rating system

No state actor builds trust between households and enterprises. A mutual rating mechanism (enterprises rate households, households rate enterprises) is the central differentiating feature.

User Flows - 3 Critical Journeys

Friction points were identified at three moments across all journeys: payment, household validation, and the USSD reporting flow.

Journey 1 - Household Discovery (web/USSD) → Registration & validation → Waste reporting → Waiting/matching → Collection by company → Remuneration

Journey 2 - Enterprise Platform registration → Household search → Selection (zone/quantity) → Platform payment → Transport & collection → Household rating

Journey 3 - Administrator Household validation → Availability monitoring → Transaction management → Revenue redistribution → Moderation & rating → Platform updates

Design Decisions

Based on research and coaching insights, the following UX solutions were designed covering the homepage, onboarding for both profiles, and the household dashboard.

User Testing Protocol

Participants: 3–5 (households and company representatives)

Duration: 30 min per session

#	Task	Focus Area
T1	You are a household. Register and declare 5 kg of available plastic waste.	Household onboarding
T2	You are a company. Find a household in Ouidah with at least 8 kg available and place an order.	Enterprise search & order
T3	Find your transaction history and check how much you earned this month.	Dashboard navigation
T4	Rate a household after completing a collection.	Post-collection rating
T5	Report that you gave 3 kg of waste to a company — update your availability.	Real-time stock update

Test Report - Key Findings

T1 — Household Registration Success: 3/5	Users didn't distinguish between 'report' and 'declare available' — both terms coexisted. Recommendation: Single button 'Add my waste', clear and unified label.
T2 — Enterprise Order Success: 2/5	Companies couldn't find the zone filter and looked for a search bar. Recommendation: Add a map with zone filters visible from the household list.
T3 — History & Earnings Success: 5/5	No issues. The 'My Earnings' tab was found easily. Recommendation: Keep current placement.
T4 — Household Rating Success: 4/5	One user didn't know rating was possible post-collection — no notification received. Recommendation: Automatic notification after each confirmed collection.

Next Steps

- Build high-fidelity prototype based on validated wireframes
- Conduct usability tests with real households and enterprises in Ouidah
- Finalise USSD flow design for offline-first household access
- Define platform launch criteria: minimum households and enterprises per zone